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Problem 1. $\pi_{\theta}(a|s) = \frac{\exp(\phi(s,a)^T \theta)}{\sum_{a'} \exp(\phi(s,a')^T \theta)}$.

$$\begin{aligned} \therefore \log \pi_{\theta}(a|s) &= \log(\exp(\phi(s,a)^T \theta)) - \log(\sum_{a'} \exp(\phi(s,a')^T \theta)) \\ &= \phi(s,a)^T \theta - \log(\sum_{a'} \exp(\phi(s,a')^T \theta)) \end{aligned}$$

$$\begin{aligned} \therefore \nabla_{\theta} \log \pi_{\theta}(a|s) &= \frac{\phi(s,a) \cdot \exp(\phi(s,a)^T \theta)}{\sum_{a'} \exp(\phi(s,a')^T \theta)} \\ &= \phi(s,a) - \sum_{a'} \phi(s,a') \pi_{\theta}(a'|s) \end{aligned}$$

$$= \phi(s,a) - \mathbb{E}_{a' \sim \pi_{\theta}(\cdot|s)} [\phi(s,a')]$$

$$\therefore \nabla_{\theta} \log \pi_{\theta}(a|s) = \phi(s,a) - \mathbb{E}_{a' \sim \pi_{\theta}(\cdot|s)} [\phi(s,a')].$$

Problem 2.

We assumed $N=1$. $\therefore i$ can be ignored.

$$G_t = \sum_{t'=t}^T r(s_{t'}, a_{t'})$$

$$\begin{aligned} \therefore \left. \begin{aligned} G_1 &= r(s_1, a_1) + \dots + r(s_t, a_t) \\ G_2 &= r(s_2, a_2) + \dots + r(s_t, a_t) \\ &\vdots \\ G_T &= r(s_t, a_t) \end{aligned} \right\} \begin{aligned} &T + (T-1) + (T-2) + \dots + 1 \\ &= \frac{T(T+1)}{2} \end{aligned} \end{aligned}$$

$$\therefore O(T^2)$$

But if we compute G_T first and G_{T-1} next, $\dots$ G_1 last.

Then we can use previous result G_{i+1} to compute G_i .

$$G_i = r(s_i, a_i) + G_{i+1}.$$

∴ We can calculate $G_1, G_2, \dots, G_T$ with $O(T)$ complexity.

The pseudocode:

Input: $T, r(s_1, a_1), r(s_2, a_2), \dots, r(s_T, a_T)$

Output: Array G which have length T and $G[i] = G_i$.

for i in range(T):

$j = T - i$

if $j = T$:

$G[j] = r(s_T, a_T)$

else:

$G[j] = G[j+1] + r(s_j, a_j)$

return G

Problem 3.

The new gradient: $\nabla_{\theta} J(\theta) = E_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(s_t, a_t) (G_t - b(s_t, a_t))]$

If it is unbiased,

$\nabla_{\theta} J(\theta)$ should be $E_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(s_t, a_t) G_t]$.

$E_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(s_t, a_t) (G_t - b(s_t, a_t))]$

$= E_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(s_t, a_t) G_t - \nabla_{\theta} \log \pi_{\theta}(s_t, a_t) b(s_t, a_t)]$

$= E_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(s_t, a_t) G_t] - E_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(s_t, a_t) b(s_t, a_t)]$.

∴ If $E_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(s_t, a_t) b(s_t, a_t)]$ is always 0,

then it is unbiased.

$$E_{\pi_{\theta}} [\nabla_{\theta} \log \pi_{\theta}(s_t, a_t) b(s_t, a_t)] = \sum_{a_t} \pi_{\theta}(a_t | s_t) \nabla_{\theta} \log \pi_{\theta}(a_t | s_t) b(s_t, a_t)$$

$$= \sum_{a_t} \nabla_{\theta} \pi_{\theta}(a_t | s_t) b(s_t, a_t) = \nabla_{\theta} \sum_{a_t} \pi_{\theta}(a_t | s_t) b(s_t, a_t)$$

If $b(s_t, a_t)$ is not constant and depend on a_t then the above value is

not 0. ∴ It is biased gradient estimate.